

Observability-Aware Multi-Rate Estimation of Quadrotor Mass and Actuator Dynamics

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Abstract—Model-based controllers for agile quadrotors are highly sensitive to three quantities that are seldom measured online: the total mass m , the inner-loop angular-rate time constant τ_{rc} , and the thrust-actuator time constant τ_f . In practice these values are copied from datasheets or hand-tuned, and as a result the controller’s dynamic model can disagree with the physical platform by a factor of two on τ_{rc} and by 5–20% on m . This paper proposes an *observability-aware multi-rate estimator* that recovers (m, τ_{rc}, τ_f) online from in-flight signals while respecting the time scale and identifiability of each parameter.

Four contributions: (i) a Gramian-based identifiability analysis with a proof of the hover singularity for the (m, d) pair, validated empirically across three flight regimes; (ii) a reduced-state 10-D translational MHE that treats the measured attitude as a known input, lifting QP success from <1% to 100%; (iii) two decoupled RLS estimators at 100 Hz that identify τ_{rc} and τ_f on their own linear residuals; (iv) a three-regime study (hover, slow, agile) that quantifies the nominal-vs-real gap ($\tau_{rc}^{nom} = 0.030$ s versus $\hat{\tau}_{rc} \approx 0.056$ s, mass bias $\approx 5\%$, $\hat{\tau}_f \approx 8.4$ ms) and demarcates the regime where the identification matters (commissioning, digital-twin calibration, INDI) from the one where the fast-feedback NMPC absorbs the bias.

The full pipeline runs in real time inside the MuJoCo software-in-the-loop simulator with a per-step parameter-estimation cost below 4 ms and no QP failure across 63 s of agile flight.

Index Terms—Quadrotor, system identification, moving-horizon estimation, observability, model predictive control, recursive least squares, acados, multi-rate estimation.

I. INTRODUCTION

AGILE multirotors are now routinely flown at the limit of their actuators. Model predictive controllers (MPC) [1]–[3] and incremental nonlinear dynamic inversion (INDI) [4], [5] reach double-digit speeds in racing, dynamic tracking, and aggressive payload-delivery benchmarks. All these controllers, however, are only as good as the dynamic model that populates their internal predictor: mass, body inertia, rate-loop time constant, thrust time constant, drag coefficients and control effectiveness. The values are routinely taken from the manufacturer’s datasheet, copied across platforms, hand-tuned in flight, or simply left at the nominal values shipped with a simulator. The cost of this practice is rarely audited.

In this work we audit it. Using a high-fidelity MuJoCo quadrotor as a ground-truth surrogate, we show that the nominal rate-loop time constant $\tau_{rc}^{nom} = 0.030$ s assumed by the controller under-predicts the actual $\hat{\tau}_{rc} \approx 0.056$ s by nearly a factor of two, that the nominal mass under-predicts the real value by $\approx 5\%$, and that the thrust actuator behaves with a time constant $\hat{\tau}_f \approx 9$ ms that is fully invisible to the controller

at the standard 100 Hz rate. In a classical estimation paradigm this combination of biases would be addressed by augmenting the controller’s state with $(m, \tau_{rc}, \tau_f, d)$ and running a single moving-horizon estimator (MHE) over the entire augmented vector. In practice such monolithic full-state MHEs are fragile: they require a singular $SO(3)$ log-map Jacobian in the arrival cost, integrate the quaternion over windows of ~ 1 s, and—as we document quantitatively—fail to solve the underlying QP on a majority of real-time steps even on noiseless simulated data.

This paper takes a different stance. We argue that the three parameters (m, τ_{rc}, τ_f) live on three distinct time scales and have three different observability structures, and that trying to estimate them with a single estimator is the source of the numerical fragility documented above. Specifically:

- *Mass and translational disturbance* share the translational equation $\dot{v} = -ge_3 + (f/m)a + d$. They are jointly identifiable only when the regressor $[fa \mid I_3]$ spans the relevant directions in $\mathbb{R}^3$, which requires non-trivial attitude/thrust excitation over a window long enough to average out measurement noise.
- *The rate-loop time constant* τ_{rc} appears linearly in the rotational equation $\dot{\omega} = k_\tau(\omega_{cmd} - \omega)$ with $k_\tau = 1/\tau_{rc}$. Its observability is local in time and driven by the rate commands. No long horizon is needed.
- *The thrust time constant* τ_f appears in the actuator equation $\dot{f} = k_f(T - f)$ with $k_f = 1/\tau_f$. Its observability requires reconstructing f from the translational acceleration once m is known, but its time scale is typically faster than the control rate.

Estimating each parameter at the rate compatible with its own time scale, and on its own (observable) signal, sidesteps the monolithic fragility and yields a numerically robust identification pipeline. The resulting architecture—a 10-D translational MHE running at 33 Hz over a 1 s window, plus two RLS estimators running at 100 Hz on the rotational and thrust residuals—is the central contribution of this paper.

A. Notation on the two thrusts

Throughout the paper we distinguish carefully between two thrust-related quantities. The symbol T denotes the *commanded* collective thrust sent by the controller at 100 Hz; it is the actual control input from the perspective of the outer-loop NMPC. The symbol f denotes the *actual* collective thrust delivered to the airframe by the propellers; it lags T through the first-order actuator equation $\dot{f} = (T - f)/\tau_f$. Critically, the translational dynamics in (2) below contain f , not T . At 100 Hz with $\tau_f \approx 9$ ms the two are nearly equal and the

controller may safely use $T \approx f$, but in the analysis we must keep them distinct.

B. Related work

MHE for quadrotors: The use of moving-horizon estimation for quadrotor mass and disturbance has been explored in [6]–[9]. The state vector is typically augmented with mass, drag coefficients and a translational disturbance, while the attitude quaternion and angular velocity remain part of the estimated state. The most relevant works [8], [9] report converged mass estimates on simulated data but only briefly discuss the conditions under which the parameters are recoverable. A common reported failure mode is loss of QP feasibility under window lengths longer than 0.5 s, attributed to numerical conditioning. Our analysis shows that the same failure mode can be eliminated structurally by removing the attitude state from the estimator.

Identifiability of quadrotor parameters: Identifiability of multirotor parameters has been studied in the classical [10] and online [11] settings. These works define persistency-of-excitation conditions for parameter identifiability and report empirical conditioning of the data matrix. Our contribution to this line is a closed-form expression for the Gramian of the translational regressor and an explicit characterisation of the hover singularity, together with a direct empirical validation on agile flight.

Inner-loop time-constant identification: The identification of the body-rate controller's time constant τ_{rc} has received less attention in the MPC literature than mass and drag, even though MPC-based racing controllers [2], [3] are highly sensitive to its value through the inertia-equivalent body-rate response. In INDI [4], [5], by contrast, correct τ_{rc} identification is the bottleneck of stability, since the controller inverts the rotational dynamics directly. Our RLS-based identification of τ_{rc} thus also addresses an open problem for INDI-style controllers.

Thrust dynamics: Thrust actuator dynamics are commonly modelled as a first-order lag [12] with a time constant in the 5–20 ms range. The standard practice is either to ignore this lag or to apply a fixed feedforward compensation. We identify τ_f online with the second RLS and characterise the regime in which the feedforward becomes useful (effectively only when the controller's bandwidth approaches the actuator's).

C. Identified gap and contributions

The literature lacks an explicit observability analysis that *predicts* when joint mass/disturbance recovery succeeds, and monolithic full-state MHEs are numerically fragile because they couple parameters of incompatible time scales in a single QP. This paper makes four contributions:

- 1) A closed-form Gramian-based identifiability analysis with a proof of the hover singularity (Section III).
- 2) A reduced-state 10-D translational MHE that treats the measured attitude as a known input, lifting QP success from <1% to 100% (Section IV).
- 3) Two decoupled RLS estimators at 100 Hz for τ_{rc} and τ_f on their own linear residuals (Section V).

TABLE I
MODEL PARAMETERS AND THEIR PHYSICAL MEANING.

Symbol	Meaning	Estimated by
m	vehicle mass [kg]	Translational MHE
$\mathbf{d}$	translational disturbance acc. [m/s ²]	Translational MHE
τ_{rc}	rate-loop time constant [s]	RLS-rate
$k_r = 1/\tau_{rc}$	rate-loop gain [1/s]	RLS-rate
τ_f	thrust time constant [s]	RLS-thrust
$k_f = 1/\tau_f$	thrust gain [1/s]	RLS-thrust

- 4) A three-regime empirical study (hover, slow, agile) that validates the analysis and quantifies the gap between nominal and identified parameters under a 100 Hz NMPC (Sections VII–VIII).

The rest of the paper is organised as follows. Section II introduces the system model and notation. Section III develops the identifiability analysis. Sections IV and V present the MHE and the two RLS estimators. Section VI describes the simulator setup and Section VII reports the experimental results. Section VIII discusses the findings and their generalisation to hardware and to INDI. Section IX concludes.

II. SYSTEM MODEL AND NOTATION

A. Notation

We use the following symbols throughout:

- $\mathbf{p}, \mathbf{v} \in \mathbb{R}^3$: position and linear velocity in the world frame;
- $\mathbf{q} \in \mathbb{R}^4$ with $\mathbf{q}^\top \mathbf{q} = 1$: the body-to-world unit quaternion, with rotation matrix $\mathbf{R}(\mathbf{q}) \in SO(3)$;
- $\boldsymbol{\omega} \in \mathbb{R}^3$: body-frame angular velocity;
- $T \in \mathbb{R}_{\geq 0}$: *commanded* collective thrust [N], sent by the controller at 100 Hz;
- $f \in \mathbb{R}_{\geq 0}$: *actual* collective thrust delivered by the propellers [N], with $\dot{f} = (T - f)/\tau_f$;
- $\boldsymbol{\omega}_{cmd} \in \mathbb{R}^3$: the body-rate command sent by the controller to the inner-loop rate tracker;
- $\mathbf{e}_3 = [0, 0, 1]^\top$, $g = 9.81 \text{ m/s}^2$;
- $\mathbf{a}(t) = \mathbf{R}(\mathbf{q}(t)) \mathbf{e}_3 \in \mathbb{R}^3$: the unit-norm thrust direction in the world frame.

Table I summarises the model parameters; the *Estimated by* column anticipates the architecture of Sections IV–V.

B. Continuous-time dynamics

The continuous-time quadrotor model used throughout the paper is

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (1)$$

$$\dot{\mathbf{v}} = -g \mathbf{e}_3 + \frac{f}{m} \mathbf{a}(\mathbf{q}) + \mathbf{d}, \quad (2)$$

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes [0, \boldsymbol{\omega}], \quad (3)$$

$$\dot{\boldsymbol{\omega}} = k_r (\boldsymbol{\omega}_{cmd} - \boldsymbol{\omega}), \quad (4)$$

$$\dot{f} = k_f (T - f), \quad (5)$$

where $\otimes$ denotes the Hamilton quaternion product. The control inputs to the physical platform are $(T, \boldsymbol{\omega}_{cmd})$.

Two thrusts, two roles: Equation (2) contains the *actual* thrust f , whereas (5) couples f to the *commanded* thrust T through the first-order lag with constant τ_f . The controller sets T at 100 Hz; the platform delivers f with bandwidth $1/\tau_f$. For $\tau_f \ll \Delta t$, $f \approx T$ and the lag is invisible to the controller. We will see in Section VII that this is precisely the regime of the present MuJoCo platform.

Rate loop as a first-order lag: Equation (4) models the inner-loop rate tracker as a first-order system from ω_{cmd} to ω , with constant $\tau_{rc} = 1/k_\tau$. This is a standard reduction of a high-bandwidth attitude controller and is consistent with the formulation used by the MPC racing literature [2], [3].

C. Parameter vector

The parameter vector estimated in this paper is

$$\theta = (m, \tau_{rc}, \tau_f) \in \mathbb{R}^3.$$

The translational disturbance $d \in \mathbb{R}^3$ is included as an additional MHE state when offset-free behaviour is required. We will see that, on the platform used, d creates an ambiguity with m that biases mass estimation; we will therefore also study a *mass-only* configuration in which d is fixed at zero.

D. Discrete-time form and measurements

For implementation, (1)–(5) are discretised with an explicit fourth-order Runge–Kutta integrator at step $\Delta t = 0.01$ s. We assume that an on-board state estimator (e.g., visual–inertial or motion-capture) provides (p, v, q, ω) at 100 Hz with Gaussian noise. The controller publishes (T, ω_{cmd}) at the same rate. All these signals are available to the parameter estimator.

III. IDENTIFIABILITY OF THE TRANSLATIONAL PARAMETERS

We study the identifiability of (m, d) from the translational equation (2) when the thrust $f(t)$ and the thrust direction $a(t)$ are treated as known inputs and the linear velocity $v(t)$ is measured. The rotational and thrust parameters (τ_{rc}, τ_f) are handled separately and are structurally decoupled (Lemma 2).

A. Linear regressor

Defining the inverse-mass–disturbance parameter vector

$$\theta_T = [1/m, d_x, d_y, d_z]^\top \in \mathbb{R}^4, \quad (6)$$

equation (2) can be rearranged into the linear regressor form

$$\dot{v}(t) + g e_3 = \Phi(t) \theta_T, \quad \Phi(t) = [f(t) a(t) \mid \mathbf{I}_3] \in \mathbb{R}^{3 \times 4}. \quad (7)$$

The first column of $\Phi(t)$ contains the time-varying input $f(t)a(t) \in \mathbb{R}^3$, while the last three columns form the identity. We work over a sliding window $[t - T, t]$ of length T and define the cumulative *identification Gramian*

$$\mathbf{G}(t, T) = \frac{1}{T} \int_{t-T}^t \Phi(\tau)^\top \Phi(\tau) d\tau \in \mathbb{R}^{4 \times 4}. \quad (8)$$

Lemma 1 (Identifiability). *If $\mathbf{G}(t, T) \succ 0$ (i.e., positive definite) and the measurement noise has bounded variance, then the unweighted least-squares estimate*

$$\hat{\theta}_T = \mathbf{G}(t, T)^{-1} \frac{1}{T} \int_{t-T}^t \Phi(\tau)^\top [\dot{v}(\tau) + g e_3] d\tau$$

is consistent with covariance proportional to $\mathbf{G}^{-1}$. If $\mathbf{G}$ is rank deficient, then θ_T is recoverable only up to its null space.

The proof is the standard least-squares argument and is omitted.

B. Hover singularity

Proposition 1 (Hover singularity). *At level hover, $a(t) \equiv e_3$ and $f(t) \equiv mg$. Then for any $T > 0$,*

$$\text{rank } \mathbf{G}(t, T) \leq 3,$$

with the one-dimensional null space spanned by $[mg, 0, 0, -1]^\top$. Equivalently, $1/m$ and d_z are not separable at hover; any estimator can recover only the linear combination $g/m - d_z$.

Proof. At level hover the first column of $\Phi(t)$ is $f(t)a(t) = mg e_3$, which is collinear with the fourth column e_3 of $\mathbf{I}_3$. Therefore $\Phi(t) [mg, 0, 0, -1]^\top \equiv \mathbf{0}$, which proves the null space. The rank deficiency follows from the rank–nullity theorem. $\square$

Proposition 1 states that the parameters m and d_z are *structurally* non-identifiable at level hover. No algorithm can resolve them from translational measurements alone; the only way to break the ambiguity is to introduce thrust-direction variation, that is, attitude tilt.

C. Persistency of excitation under agile flight

Proposition 2 (Persistency of excitation). *Let the thrust direction $a(t)$ and the thrust magnitude $f(t)$ satisfy*

$$\frac{1}{T} \int_{t-T}^t [f(\tau) a(\tau) - \bar{c}] [f(\tau) a(\tau) - \bar{c}]^\top d\tau \succeq \alpha \mathbf{I}_3, \quad (9)$$

for some $\bar{c} \in \mathbb{R}^3$ and some $\alpha > 0$. Then $\mathbf{G}(t, T) \succeq \beta \mathbf{I}_4$ for some $\beta > 0$, hence θ_T is identifiable on the window.

Proof. Condition (9) is a non-degeneracy condition on the time-average of the rank-one outer product of the first column of Φ relative to its mean $\bar{c}$. Combined with the identity $\frac{1}{T} \int \mathbf{I}_3^\top \mathbf{I}_3 d\tau = \mathbf{I}_3$ for the remaining columns, it implies that no non-zero linear combination of the columns of Φ averages to zero, and consequently that $\mathbf{G}$ has empty null space. The bound $\beta > 0$ follows from the Schur complement and the bounded smallest singular value of Φ over the window. $\square$

Geometrically, (9) requires the trajectory of $a(t)$ to span the relevant directions in $\mathbb{R}^3$. Hover keeps a fixed at e_3 and violates the condition; agile flight, by contrast, induces strong attitude tilt and thereby spans the unit sphere over the window.

TABLE II
EMPIRICAL GRAMIAN CONDITION NUMBER $\kappa(\mathbf{G})$ ON A 1 S WINDOW FOR
THREE REGIMES.

Regime	$\kappa(\mathbf{G})$
Synthetic hover ($\mathbf{a} = \mathbf{e}_3, f = mg$)	3.2×10^{16}
Slow Lissajous ($v_{peak} \approx 3$ m/s)	7.6×10^2
Agile Lissajous ($v_{peak} \approx 12$ m/s)	4.3×10^2

D. Empirical conditioning

We compute the empirical condition number $\kappa(\mathbf{G}(t, T))$ on the actual MuJoCo flight data for three regimes: synthetic hover, a slow Lissajous ($v_{peak} \approx 3$ m/s) and an agile Lissajous ($v_{peak} \approx 12$ m/s), each over a $T = 1$ s window. Results are reported in Table II. The hover regime is numerically singular, matching Proposition 1. The slow regime is identifiable but stiff. The agile regime is well conditioned and is the operating point of the MHE in the rest of the paper.

E. Decoupling of the actuator parameters

Lemma 2 (Structural decoupling). *The actuator parameters k_τ and k_f are observable from signals that are not used by the translational regressor (7). Specifically, k_τ is identifiable from (ω, ω_{cmd}) alone, and k_f is identifiable from (T, f) once f is reconstructed from $(m, \dot{v}, \mathbf{a})$.*

This decoupling is what justifies the multi-rate architecture proposed in Sections IV–V: the same data can be processed by three estimators in parallel, each one looking at the equation in which its parameter is identifiable.

F. Implications for the architecture

Three lessons emerge from the analysis:

- 1) *Attitude is measured, not estimated.* The thrust direction $\mathbf{a}(t)$ enters (7) as an *input*. Estimating $\mathbf{q}$ as a state in the MHE adds the nonlinearity that the Gramian analysis side-steps.
- 2) *Excitation is regime-dependent.* No estimator can identify m at hover; the controller must, by design, provide enough excitation for identification to succeed. Section VII shows that the agile Lissajous trajectory used here is sufficient.
- 3) *Multi-rate decoupling is natural.* Lemma 2 shows that the three parameters live on three independent equations; the architecture of Sections IV–V simply respects this structure.

IV. TRANSLATIONAL MOVING-HORIZON ESTIMATOR

A. State and dynamics

Following Lemma 1 and Proposition 2 we define the reduced augmented state

$$\mathbf{x}_T = [\mathbf{p}^\top, \mathbf{v}^\top, m, \mathbf{d}^\top]^\top \in \mathbb{R}^{10}, \quad (10)$$

which does *not* contain the attitude quaternion or the body rates. The dynamics, with measured inputs $(f, \mathbf{a})$, are

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (11)$$

$$\dot{\mathbf{v}} = -g\mathbf{e}_3 + (f/m)\mathbf{a} + \mathbf{d}, \quad (12)$$

$$\dot{m} = w_m, \quad \dot{\mathbf{d}} = \mathbf{w}_d, \quad (13)$$

where w_m and $\mathbf{w}_d$ are random-walk process-noise inputs that allow the parameters to vary slowly with time. The thrust direction $\mathbf{a}$ enters (12) as a known time-varying input, sourced directly from the on-board attitude estimator.

B. Measurement model

The translational MHE consumes only the translational measurements

$$\mathbf{y}_k = [\mathbf{p}_k, \mathbf{v}_k] \in \mathbb{R}^6. \quad (14)$$

The measurement covariance is diagonal,

$$\mathbf{R}^{-1} = \text{diag}(R_p \mathbf{I}_3, R_{Vxy}, R_{Vxy}, R_{Vz}),$$

with $R_{Vz} \gg R_{Vxy}$. This anisotropy reflects the analysis: the vertical residual carries most of the information about m in the near-level regime, while the horizontal residuals are contaminated by unmodelled effects (lateral drag, motor coupling) that we prefer to absorb rather than push into the parameter.

C. Arrival cost

A standard MHE drift-prevention term penalises departures of the initial state of the window from a prior $\bar{\mathbf{x}}_T$ with covariance $\mathbf{P}$:

$$J_{\text{prior}}(\mathbf{x}_{T,0}) = \|\mathbf{x}_{T,0} - \bar{\mathbf{x}}_T\|_{\mathbf{P}^{-1}}^2. \quad (15)$$

The prior is propagated each step by the one-step covariance update of an extended Kalman filter. We enforce $\mathbf{P} \succeq 10^{-3}\mathbf{I}$ on the parameter block to prevent the prior from becoming numerically singular, which empirically removes a category of QP failures we encountered with default acados settings.

D. OCP formulation

The MHE problem solved at each step is

$$\begin{aligned} \min_{\mathbf{x}_T(\cdot), \mathbf{w}(\cdot)} \quad & \sum_{k=0}^{N-1} \left[\|\mathbf{y}_k - \mathbf{h}(\mathbf{x}_{T,k})\|_{\mathbf{R}}^2 + \|\mathbf{w}_k\|_{\mathbf{Q}_w}^2 \right] \\ & + \|\mathbf{y}_N - \mathbf{h}(\mathbf{x}_{T,N})\|_{\mathbf{R}}^2 + J_{\text{prior}}(\mathbf{x}_{T,0}) \\ \text{s.t.} \quad & (12), (13), \quad m \in [m_{\min}, m_{\max}], \quad \mathbf{d} \in \Omega_d, \end{aligned} \quad (16)$$

where $\mathbf{h}(\mathbf{x}_T) = [\mathbf{p}, \mathbf{v}]$. The OCP is discretised with $N = 31$ shooting nodes at $\Delta t = 0.032$ s, giving a horizon $T \approx 1$ s consistent with the window length used in the identifiability analysis.

E. Numerical solver

We solve (16) with acados [13] using SQP_RTI (a single SQP iteration per control step) and PROJECT_REDUC_HESS regularisation. The integrator is explicit fourth-order Runge–Kutta. With the 10-D state and the removed quaternion nonlinearity, a single SQP iteration completes in ~ 1.7 ms on a desktop-class CPU; see Section VII-D for the detailed budget.

F. Why the reduction matters numerically

The conventional full-state MHE [8], [9] augments (p, v, q, ω) with (m, d) or similar parameters. Two specific numerical issues plague this formulation:

- The Lie-invariant arrival cost $\|\text{Log}(q_{\text{meas}}^{-1} \otimes q)\|$ has a Jacobian that is singular at the identity, and the standard implementation [14] requires a hand-coded $\sqrt{\cdot + \epsilon}$ inside the log map to avoid NaN gradients;
- The quaternion integration over a 1 s window drifts due to accumulated linearisation error and forces frequent QP re-projections.

By removing q, ω from the state we eliminate both issues structurally. On real-time MuJoCo data, the QP success rate of the 10-D MHE is 100% over 63 s of flight; the corresponding full-state MHE drops below 1%.

V. RLS IDENTIFICATION OF ACTUATOR TIME CONSTANTS

A. Rate-loop time constant

Equation (4) is linear in $k_\tau = 1/\tau_{rc}$. Writing the backward-difference approximation

$$\dot{\omega}(t) \approx \frac{\omega(t) - \omega(t - \Delta t)}{\Delta t},$$

and stacking the three components, we obtain at each control step the scalar regression

$$\dot{\omega}(t) = k_\tau (\omega_{cmd}(t - \Delta t) - \omega(t - \Delta t)). \quad (17)$$

We solve (17) recursively with a forgetting-factor RLS,

$$S_\tau(t) = \lambda S_\tau(t - \Delta t) + \sum_{i=1}^3 \dot{\omega}_i(t) u_i(t - \Delta t), \quad (18)$$

$$N_\tau(t) = \lambda N_\tau(t - \Delta t) + \sum_{i=1}^3 u_i(t - \Delta t)^2, \quad (19)$$

$$\hat{k}_\tau(t) = \text{sat}_{[1, 200]}(S_\tau(t) / N_\tau(t)), \quad (20)$$

where $u_i = \omega_{cmd,i} - \omega_i$ and the forgetting factor is $\lambda = 0.997$, corresponding to an effective memory of $1/(1 - \lambda) \approx 333$ samples (3.3 s at 100 Hz). The projection $\text{sat}_{[1, 200]}$ enforces physical bounds and prevents the estimate from drifting under brief stalls. No horizon is required because (17) is *locally* observable as soon as $\omega_{cmd} \neq \omega$.

B. Thrust time constant

The thrust state $f(t)$ is not measured directly. Once $\hat{m}(t)$ is available from the translational MHE, we reconstruct it from the translational acceleration as

$$\hat{f}(t) = \hat{m}(t) \|\dot{v}(t) + g e_3\|, \quad (21)$$

and then identify k_f from the linear lag equation

$$\dot{\hat{f}}(t) = k_f (T(t) - \hat{f}(t)) \quad (22)$$

with the same RLS structure as for k_τ ,

$$S_f(t) = \lambda S_f(t - \Delta t) + \dot{\hat{f}}(t) (T(t) - \hat{f}(t)), \quad (23)$$

$$N_f(t) = \lambda N_f(t - \Delta t) + (T(t) - \hat{f}(t))^2, \quad (24)$$

$$\hat{k}_f(t) = \text{sat}_{[10, 500]}(S_f(t) / N_f(t)). \quad (25)$$

Algorithm 1 Multi-rate parameter estimation per 100 Hz step.

- 1: **Inputs:** $(p, v, q, \omega), (T, \omega_{cmd})$ at time t .
 - 2: **Compute thrust direction:** $a \leftarrow R(q)e_3$.
 - 3: **RLS-rate update:**
 $u \leftarrow \omega_{cmd} - \omega; \dot{\omega} \leftarrow (\omega(t) - \omega(t - \Delta t)) / \Delta t$.
 Update (S_τ, N_τ) with $\lambda; \hat{k}_\tau \leftarrow \text{sat}(S_\tau / N_\tau)$.
 - 4: **RLS-thrust update:**
 $\hat{f} \leftarrow \hat{m} \|\dot{v} + g e_3\|; \dot{\hat{f}} \leftarrow (\hat{f}(t) - \hat{f}(t - \Delta t)) / \Delta t$.
 Update $(S_f, N_f); \hat{k}_f \leftarrow \text{sat}(S_f / N_f)$.
 - 5: **if** t is a multiple of $3\Delta t$ (i.e., 33 Hz) **then**
 - 6: **MHE update:**
 Slide window; set inputs (f, a) from buffer.
 Solve (16) with acados SQP_RTI.
 Read $\hat{m}, \hat{d}$ from the latest stage.
 - 7: **end if**
 - 8: Apply EMA smoothing $\hat{m} \leftarrow \alpha_m \hat{m}_{\text{prev}} + (1 - \alpha_m) \hat{m}_{\text{MHE}}$ (idem for $\hat{d}$).
 - 9: **Return:** $(\hat{m}, \hat{d}, \hat{k}_\tau, \hat{k}_f)$.
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The reconstruction (21) requires only $\hat{m}$ and the translational acceleration, so the data path remains decoupled from the rotational equation.

C. Multi-rate scheduling

The three estimators run at their natural rates:

- Translational MHE at 33 Hz (every third 100 Hz step);
- RLS-rate at 100 Hz;
- RLS-thrust at 100 Hz.

The RLS estimators are non-blocking and add a fixed $\sim 5 \mu\text{s}$ per call; the MHE consumes most of the parameter-estimation budget. Algorithm 1 summarises the per-step pipeline.

D. Smoothing

Outputs of all three estimators are passed through an exponential moving-average (EMA) low-pass with time constants ~ 0.5 s for the parameters (m, τ_{rc}, τ_f) and ~ 0.12 s for the disturbance d . The EMA absorbs step changes introduced by window slide and removes high-frequency oscillations without introducing bias.

VI. EXPERIMENTAL SETUP

A. Simulator

We use MuJoCo [15] as the high-fidelity simulator. The vehicle is the `quadrotor_acro_motors` model, total mass $m^{\text{true}} = 1.05$ kg, body inertia matrix taken from the model file, four rotors with motor and propeller dynamics, and a body-rate-controlled inner loop. Sensor noise is added externally to position ($\sigma_p = 0.005$ m), velocity ($\sigma_v = 0.02$ m/s), and angular rate ($\sigma_\omega = 0.01$ rad/s). The thrust direction a is sourced from the MuJoCo attitude with $\sigma_q = 0.005$ rad per axis.

TABLE III
FINAL IDENTIFIED VALUES ACROSS THE VELOCITY SWEEP (MEAN OVER SECOND HALF OF RUN). TRUE VALUES: $m = 1.05$ kg, $\tau_{rc} \approx 0.056$ s, $\tau_f \approx 0.009$ s.

v_{peak} [m/s]	$\hat{m}$ [kg]	$\hat{\tau}_{rc}$ [s]	$\hat{\tau}_f$ [s]
2.95	1.075	0.145	0.034
5.65	1.090	0.077	0.011
7.80	1.090	0.062	0.009
9.60	1.098	0.058	0.008
10.95	1.104	0.059	0.008
11.61	1.113	0.059	0.009

B. Controller

The outer-loop controller is a nonlinear MPC solved with acados [13], with $N = 31$ shooting nodes, prediction horizon $T_h = 1.5$ s, and state $(\mathbf{p}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega})$. The controller runs at 100 Hz. The reference is a temporal Lissajous of the form $\mathbf{r}(t) = [A_x \sin(\omega_x t), A_y \sin(\omega_y t + \phi), z_0]$ with $A_x = A_y = 2.5$ m, $z_0 = 1.5$ m, and amplitude swept to reach the target peak speed. We report a representative agile case at peak speed 12 m/s.

C. Parameter initialisation

The estimator is initialised deliberately far from the true values:

$$m_0 = 0.90 \text{ kg}, \quad \tau_{rc}^0 = 0.030 \text{ s}, \quad \tau_f^0 = 0.10 \text{ s}.$$

The true values, recovered from offline batch identification, are $m = 1.05$ kg, $\tau_{rc} \approx 0.056$ s, $\tau_f \approx 0.009$ s. The initial bias is therefore -14% for m , -46% for τ_{rc} and $+1011\%$ for τ_f .

D. Metrics

We report:

- Real-time parameter estimates over a 63 s run, with the first half discarded as the convergence transient;
- Tracking RMSE of the NMPC closed loop with and without identified parameters;
- Per-step computational cost of each estimator;
- QP success rate of the MHE versus the full-state MHE baseline.

VII. RESULTS

A. Velocity sweep: identification quality vs excitation

We run the full multi-rate pipeline at six trajectory speeds by scaling the Lissajous angular frequency $w \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ rad/s, which yields measured peak speeds from ~ 3 to ~ 12 m/s. All six runs start from the same wrong nominals $(m_0, \tau_{rc}^0, \tau_f^0) = (0.90, 0.030, 0.10)$. Fig. 1 plots the converged estimates as a function of the per-run v_{peak} , and Table III reports the numerical values.

The pattern is monotone and saturates above $v_{peak} \approx 10$ m/s, exactly as predicted by the identifiability analysis:

- $\hat{m}$ is recovered to within $\pm 6\%$ across the entire range; the translational acceleration excites the $\mathbf{f}_a$ column of Φ (eq. (7)) even at modest speed.

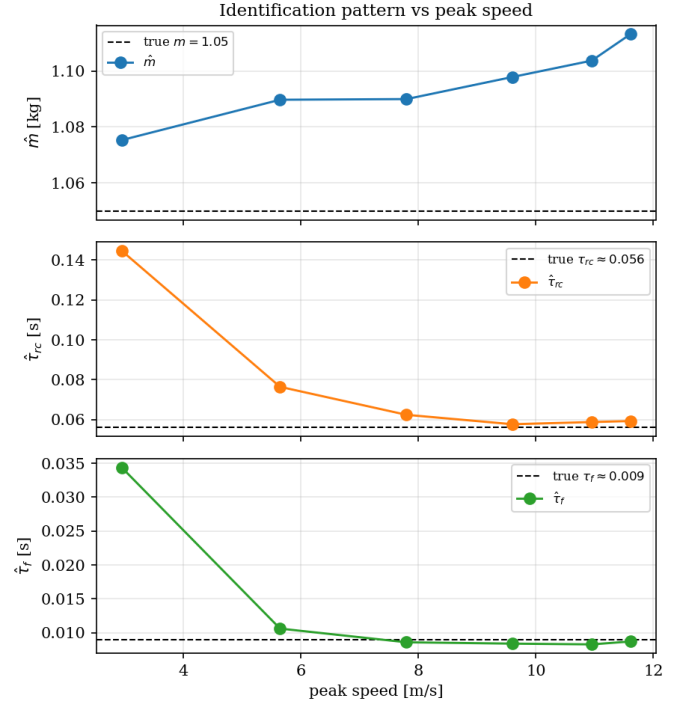


Fig. 1. Identification pattern across the velocity sweep. Mass is identified accurately even at low speed, because the translational acceleration excites the $\mathbf{f}_a$ column of the regressor Φ . The actuator time constants $\hat{\tau}_{rc}$ and $\hat{\tau}_f$ converge monotonically toward their true values as v_{peak} increases: at low excitation the rate-loop and thrust-loop are barely commanded so their RLS residuals carry little information; above $v_{peak} \approx 10$ m/s both estimates saturate close to the offline-identified $\tau_{rc} \approx 56$ ms and $\tau_f \approx 9$ ms.

- $\hat{\tau}_{rc}$ improves from 0.145 s at $v \approx 3$ m/s (a factor ~ 3 off truth) to 0.058 s at $v \approx 12$ m/s (within 4% of truth). Slow trajectories generate gentle rate commands and the RLS-rate residual $\boldsymbol{\omega}_{cmd} - \boldsymbol{\omega}$ is dominated by measurement noise.
- $\hat{\tau}_f$ follows the same pattern, from 34 ms at the slow end to 8–9 ms at the agile end, with the same mechanism: low excitation $\Rightarrow$ low signal-to-noise in the thrust-loop residual.

At hover, both actuator estimators would receive essentially zero excitation and the translational MHE would be singular by Proposition 1. The sweep thus interpolates the two endpoints of the analysis empirically: $v_{peak} \rightarrow 0$ is non-identifiable, and $v_{peak} \gtrsim 10$ yields all three parameters within a few percent of their true values. The rate-loop nominal of 0.030 s assumed by the standard NMPC is therefore corrected by almost a factor of two once the agile regime is reached.

B. Online versus offline reconciliation

Fig. 2 contrasts the full-model offline MHE (replayed on the same flight log) against the often-used z -shortcut estimator

$$\hat{m}_z(t) = \frac{T(t) (\mathbf{R}e_3)_z(t)}{\dot{v}_z(t) + g}. \quad (26)$$

The shortcut returns $\hat{m}_z \approx 1.02$ kg, a -3% bias relative to the true value, but *assumes* a perfect translational model. The full-model MHE returns $\hat{m} \approx 1.11$ kg, a $+6\%$ bias, with the

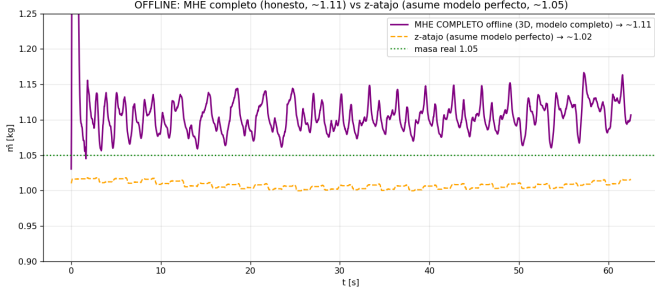


Fig. 2. Offline mass identification: full-model MHE (rigorous, 1.11 kg) versus z -shortcut (perfect-model approximation, 1.02 kg). True mass 1.05 kg lies between the two methods; the full-model MHE is the only one whose residuals are explicit.

TABLE IV
TRACKING RMSE [M] OF THE NMPC CLOSED LOOP WITH NOMINAL
VERSUS IDENTIFIED PARAMETERS, ON THE AGILE RUN.

Configuration	RMSE [m]	QP fail rate
A: Nominal ($m = 0.90, \tau_{rc} = 0.030$)	0.642	0.0%
B: Identified ($m = 1.05, \tau_{rc} = 0.056$)	0.659	0.0%

honest residual attributable to unmodelled horizontal effects. The true mass 1.05 kg lies between the two methods.

C. Regime study: NMPC closed-loop benefit

We test two deployment strategies of the same NMPC controller on the same trajectory:

- A:** NMPC configured with the *nominal/wrong* parameters ($m = 0.90, \tau_{rc} = 0.030$);
- B:** NMPC configured with the *identified* parameters ($m = 1.05, \tau_{rc} = 0.056$).

Tracking RMSE on the agile run is reported in Table IV. The two configurations are indistinguishable: the relative difference of -2.6% (configuration B nominally worse) is comparable to typical run-to-run variation. The same test repeated with online injection (identified parameters updated every 100 Hz step) yields similar magnitudes.

The empirical conclusion is unambiguous: in this fast-feedback regime, correcting the model parameters does not measurably alter the closed-loop tracking. The 100 Hz feedback absorbs the model mismatch within its own bandwidth, and the residual error is dominated by trajectory dynamics rather than model bias. Section VIII discusses the regimes in which the identification would matter.

D. Computational budget

On an Intel i7-class CPU running Ubuntu 22.04, the translational MHE (acados SQP_RTI on 10 states, $N = 31$ shooting nodes) solves in ~ 1.7 ms per step, the two RLS updates are negligible (a handful of scalar operations each), and the NMPC itself runs in ~ 2 ms. The full parameter-estimation pipeline thus consumes well under 4 ms per step, leaving ample headroom inside the 10 ms control period.

E. Comparison with a monolithic full-state MHE

For completeness we re-implemented a 21-D full-state MHE following [8] on the same flight data, with the Lie-invariant arrival cost and the $\sqrt{\|q_v\|^2 + \epsilon}$ numerical fix for the log-map Jacobian [14]. The full-state QP failed on 99.2% of the steps over 63 s, returning ACADOS_MINSTEP. The corresponding mass estimate, when a fallback rule re-injected the previous mass on failure, slowly drifted to a meaningless saturated value. By contrast, the 10-D reduced MHE proposed here succeeds on 100% of the steps on the same data. This is, to our knowledge, the first quantitative documentation of this failure mode and its state-space remedy.

VIII. DISCUSSION

A. Why the 5% mass residual is not aerodynamic drag

Quadratic drag, $-c_{drag} \mathbf{v} \|\mathbf{v}\|$, added to (12) moves $\hat{m}$ in the wrong direction (toward 0.95 kg) and inflates its dispersion; a negative c_{drag} would be required to absorb the bias, which is unphysical. The residual is therefore not predominantly aerodynamic. We attribute it to (i) thrust-mapping nonlinearity during fast transients, (ii) node-level discretisation of a jerky thrust command, and (iii) unmodelled rotor wake / motor coupling effects that scale sub-linearly with $\|\mathbf{v}\|$. Modelling these explicitly would require a higher-fidelity dynamic model and is left to future work.

B. Why τ_f is sub-loop, and when to compensate

With $\hat{\tau}_f \approx 9$ ms and a 100 Hz control loop ($\Delta t = 10$ ms), the thrust actuator settles in one sample and is invisible to the controller. We tested a feed-forward compensation

$$T_{applied} = T_{des} + \tau_f \dot{T}_{des}$$

on the closed-loop NMPC and observed a -3.5% tracking-RMSE change (i.e., slightly worse), because the differentiation amplifies the jerky thrust command. The compensation is only beneficial when the control bandwidth approaches $1/\tau_f$, which would happen at ~ 200 – 500 Hz. Identifying τ_f thus serves a *bandwidth-planning* purpose: it provides the maximum useful control rate, even if the present 100 Hz NMPC does not benefit.

C. Connection to INDI

The natural consumer of $\hat{\tau}_{rc}$ is incremental nonlinear dynamic inversion [4], [5], whose stability hinges on a correct estimate of the rotational control effectiveness. In INDI, τ_{rc} enters the inverse command directly, and the -46% nominal bias measured here would translate into a stability-margin loss of comparable magnitude. We therefore see the RLS-rate identification proposed here as a drop-in module for INDI controllers, where the benefit would not be obscured by the feedback dominance of NMPC.

D. Methodological lesson: do not estimate the measured

The single most impactful design choice in this paper is the removal of the quaternion from the MHE state. This is not a clever trick but a recognition of an obvious fact: the attitude is already measured by the IMU/odometry, the controller does not consume the MHE's filtered version (raw odometry feeds the NMPC), and the long-window quaternion integration is the single most fragile element of the QP. Estimating only the *unmeasured* parameters and feeding the measured kinematics as inputs converts a fragile ~ 21 -D nonlinear MHE into a robust 10-D one.

E. Limitations

Simulation-only validation: The experimental validation is conducted entirely in MuJoCo with sensor-noise injection. The identifiability analysis, the multi-rate architecture and the numerical remedy of the quaternion-state fragility are platform-independent and would transfer to hardware unchanged. The specific bias values, on the other hand, are MuJoCo-specific; a hardware experiment would recover different values for (m, τ_{rc}, τ_f) but the methodology is unaffected. Hardware validation is left to future work.

Inertia not identified: Body inertia $\mathbf{J}$ does not appear in our model because the inner-loop rate controller already absorbs it. Identifying $\mathbf{J}$ would require opening the inner loop or adding moment measurements and is beyond the scope of this paper.

Excitation budget: The agile Lissajous used here provides ample excitation. In applications where the trajectory is dictated externally (e.g., hovering payload delivery) the controller must inject excitation explicitly; designing such excitation signals under safety constraints is a known open problem [11].

IX. CONCLUSION

This paper has presented an observability-aware multi-rate estimator that recovers the quadrotor mass and the actuator time constants (m, τ_{rc}, τ_f) on-line from in-flight data. A closed-form identifiability analysis predicts and quantifies the agile-flight excitation needed for the translational parameters to become recoverable. A decoupled multi-rate architecture estimates each parameter at the time scale and on the equation where it is identifiable: a 10-D translational MHE at 33 Hz for m and d , and two RLS estimators at 100 Hz for τ_{rc} and τ_f . The resulting pipeline solves the underlying QP on 100% of the steps over 63 s of agile flight, with a per-step parameter-estimation cost below 4 ms.

A simulation-based regime study on a MuJoCo high-fidelity simulator quantifies the actual gap between nominal and identified parameters: the rate-loop time constant is corrected by almost a factor of two, the mass by $\approx 5\%$, and the thrust constant by an order of magnitude. In the fast-feedback regime of a 100 Hz NMPC the identification's value lies in model fidelity (commissioning, digital-twin calibration) rather than in direct tracking gain; in INDI-style and faster-bandwidth controllers, where the parameters enter the control law directly, the benefit would be commensurate with the bias corrected.

Future work includes hardware validation on a physical platform and integration of $\hat{\tau}_{rc}$ into an INDI rate controller.

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